

ARKANSAS INTL, AR, USA

WMO: 723408

Lat: 35.967N

Lon: 89.950W

Elev: 254

StdP: 14.56

Time Zone: -6.00 (NAC)

Period: 82-92

WBAN: 13814

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.7	17.4	0.5	5.7	13.7	7.0	7.9	19.3	26.9	33.1	24.1	40.3	9.3	10	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	97.0	78.4	94.7	77.4	92.5	76.7	81.5	91.7	79.9	90.3	78.5	88.9	6.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.6	150.0	87.8	77.1	142.2	86.3	75.5	134.7	84.4	45.2	91.8	43.5	90.8	42.1	89.2	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	19.1	17.2	5.2	100.1	9.1	4.1	-1.4	103.1	-6.7	105.5	-11.8	107.8	-18.5	110.7		
(5)			3.9	83.2	8.9	1.7	-2.5	84.4	-7.7	85.4	-12.7	86.4	-19.1	87.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.0	37.1	42.4	52.0	60.5	70.6	78.9	82.1	80.0	73.4	62.2	51.2	40.8		
	DBStd	17.63	9.98	10.56	9.81	8.73	6.38	5.28	4.57	4.98	7.94	7.75	9.65	12.05		
	HDD50	1206	409	249	97	16	0	0	0	0	0	5	103	327		
	HDD65	3464	864	632	415	186	24	0	0	0	26	147	419	751		
	CDD50	5234	10	37	159	330	638	867	995	931	701	383	140	43		
	CDD65	2018	0	0	11	51	197	417	530	466	277	61	7	1		
	CDH74	20539	0	2	63	393	1525	4400	6176	5009	2529	410	31	1		
	CDH80	9074	0	0	5	100	481	2015	3056	2328	1008	79	2	0		
(14) Wind	WSAvg	6.9	8.2	8.5	8.9	8.1	7.0	5.7	4.9	4.5	5.0	6.0	7.8	8.3		
(15) Precipitation	PrecAvg	51.1	4.1	4.3	4.9	5.1	5.1	4.1	4.3	2.9	3.3	4.0	4.0	5.1		
	PrecMax	64.6	8.7	8.7	10.8	11.0	10.9	7.8	10.6	5.3	6.7	10.2	9.2	12.6		
	PrecMin	38.1	1.7	1.6	1.2	1.3	0.9	1.5	0.8	0.6	0.4	0.0	1.0	0.9		
	PrecStd	7.2	1.7	1.9	2.3	2.4	2.4	1.6	2.5	1.7	1.8	2.4	2.1	2.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.5	70.7	80.3	88.1	91.2	99.2	101.5	99.0	95.3	86.3	77.7	70.3		
		MCWB	57.1	61.3	65.5	69.1	73.5	78.6	78.2	80.0	75.3	69.9	64.6	66.4		
	2%	DB	60.2	65.8	75.1	82.7	88.1	95.6	97.4	95.9	92.0	82.1	73.1	65.6		
		MCWB	55.2	57.2	62.3	67.7	73.0	76.6	78.5	78.6	75.6	68.2	64.0	62.1		
	5%	DB	56.2	62.0	71.4	78.6	85.3	92.8	94.9	93.5	89.6	78.8	70.1	62.3		
		MCWB	51.6	56.9	62.1	66.1	72.4	75.6	77.9	77.9	75.5	66.0	62.9	58.7		
	10%	DB	51.6	58.1	67.7	74.5	82.4	90.2	92.6	90.9	86.3	75.1	66.4	58.5		
		MCWB	47.2	53.2	60.1	63.8	70.1	74.5	77.3	77.2	73.7	64.0	60.5	54.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.5	65.0	70.0	72.8	77.9	81.6	83.5	83.8	80.1	73.6	68.6	66.9		
		MCDB	62.9	68.6	77.2	84.0	87.0	93.8	93.4	96.3	88.7	81.3	73.1	69.8		
	2%	WB	56.5	60.6	66.8	69.9	75.9	79.5	81.8	81.3	78.0	71.1	66.3	63.2		
		MCDB	59.2	63.2	71.5	77.8	84.3	91.8	91.5	91.9	88.1	76.9	70.3	65.5		
	5%	WB	51.8	57.7	64.0	68.1	74.2	77.9	80.1	79.6	76.5	69.2	64.1	59.7		
		MCDB	55.5	60.6	68.5	75.6	81.6	89.4	90.2	89.6	86.2	75.5	68.6	61.8		
	10%	WB	47.6	53.3	60.1	65.8	72.5	76.5	78.6	78.3	75.1	67.3	61.3	55.3		
		MCDB	51.4	57.4	66.8	73.0	79.5	86.5	89.0	88.0	83.8	73.3	66.3	57.8		
(35) Mean Daily Temperature Range	MDBR		14.4	15.4	17.5	18.6	18.1	18.9	18.6	18.8	19.9	20.0	16.4	14.9		
	5% DB	MCDBR	21.0	21.4	21.7	23.2	20.6	22.1	21.1	20.8	21.2	22.5	20.5	17.5		
		MCWBR	17.3	15.4	13.4	12.3	9.8	8.7	7.4	7.6	9.1	11.5	14.8	15.1		
	5% WB	MCDBR	18.9	17.4	17.3	19.1	17.4	19.0	18.7	18.6	18.7	16.9	17.9	15.8		
		MCWBR	17.4	14.7	13.6	11.3	9.0	8.5	8.0	8.2	9.2	9.9	15.2	15.1		
(40) Clear-Sky Solar Irradiance	taub		0.316	0.328	0.366	0.407	0.439	0.456	0.486	0.482	0.417	0.358	0.326	0.315		
	taud		2.485	2.444	2.370	2.253	2.228	2.183	2.138	2.127	2.303	2.453	2.494	2.498		
	Ebn at Noon		280	289	286	278	270	264	255	253	266	275	274	272		
	Edh at Noon		26	31	36	43	45	47	49	48	38	30	26	25		
(44) All-Sky Solar Radiation	RadAvg		748	921	1275	1642	1841	2067	1995	1827	1541	1184	828	638		
	RadStd		65	122	108	110	133	128	99	106	101	137	85	63		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page